

Diffusion Processes for Inference of Structural Causal Models

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May 31th, 2026

Inference of Structural Causal Models

The goal is to infer an SCM-model with optimal

- 1) complexity $\mathcal{L}_C$,
- 2) accuracy $\mathcal{L}_D$,
- 3) robustness $\mathcal{L}_{ci}$

The models:

- 1) the model f_{ai} maps a dataset $\mathbf{D}$ to a model $f_G(\mathbf{w}, \mathbf{x})$,
- 2) the model f_G maps a sample $\mathbf{x} \in \mathbf{D}$ to its forecast $\hat{\mathbf{x}}$, provided optimal parameters $\hat{\mathbf{w}}$.

The graph rewriting

A graph $G(V, E)$ with the

- 1) coloring function $c(V)$,
- 2) weights $\mathbf{w}(V)$,
- 3) probability of the vertices $p(V)$,
- 4) probability of the edges $\mathbf{Z} \in \{0, 1\}^{n \times n}$ or $\mathbf{\Gamma} \in [0, 1]^{n \times n}$.

The graph G defined the function $f_G(\mathbf{w}, \mathbf{x}) \rightarrow \hat{\mathbf{x}}$.

There given a graph rewriting rules $G \mapsto G'$ so that each step delivers an admissible graph (admissible means a graph G defines a function f_G).

To produce a sequence of the graphs introduce a vector $\mathbf{z} = \mathbf{vec}(\mathbf{Z})$.

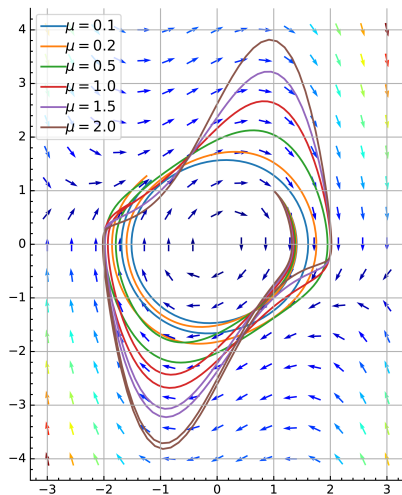
The graph probability

Introduce a Metropolis Hastings graph sampling procedure

The graph probability

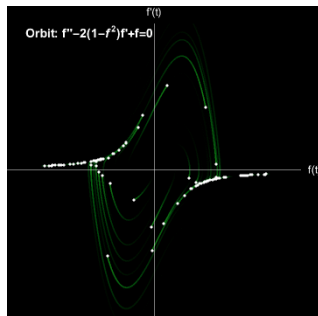
Introduce a Metropolis Hastings graph sampling procedure

Van der Pol oscillator¹

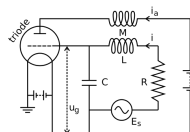


Phase portrait of van der Pol's equation $\frac{d^2y}{dt^2} + \mu(y^2 - 1)\frac{dy}{dt} + y = 0$.

¹Wikipedia: Van der Pol oscillator



Randomly chosen initial conditions are attracted to a stable orbit



Electrical circuit resulting in a forced Van der Pol oscillator

The van der Pol oscillator model

The forced Van der Pol oscillator:

$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x - A\sin(\omega t) = 0,$$

has two linear and one non-linear parameters: μ, A, ω , where A is the amplitude, of the wave function and ω is its angular velocity.

The causal graph has four vertices, the variables:

$$X_1, X_2, X_3 = \ddot{x}, \dot{x}, x$$

and the time t .

An alternative form is a time-independent Hamiltonian by augmenting it to a four-dimensional, $X_1, \dots, X_4$ autonomous dynamical system using an auxiliary second-order nonlinear differential equation:

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0, \quad \ddot{y} + \mu(1 - x^2)\dot{y} + y = 0.$$

The Hamiltonian

$$H(x, y, p_x, p_y) = p_x p_y + xy - \mu(1 - x^2) y p_y,$$

where

$$p_x = \dot{y} + \mu(1 - x^2)y \quad \text{and} \quad p_y = \dot{x}$$

are the conjugate momenta corresponding to "x" and "y".